

Qatar-2b

R-1387

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_210512 · created 2026-08-01T02:31:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459346.8001 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.101 +/- 0.035
Transit depth (Rp/Rs)^2	1.02 +/- 0.7 [%]
Orbital Inclination (inc)	83.99 +/- 1.77
Airmass coefficient 1 (a1)	0.942 +/- 0.018
Airmass coefficient 2 (a2)	0.042 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.93 %
Best Comparison Star	#2 - [144, 88]
Optimal Aperture	3.02
Optimal Annulus	9.55
Transit Duration (day)	0.055 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.101 $\pm$ 0.035 vs 0.16327 (deviation 1.78 σ)
Duration fit vs published	0.055 d vs 0.0767 d (deviation 1.67 σ)
Mid-time O-C	0.8 min
Depth SNR	2.9
Residual scatter	1.93 %

Light curve

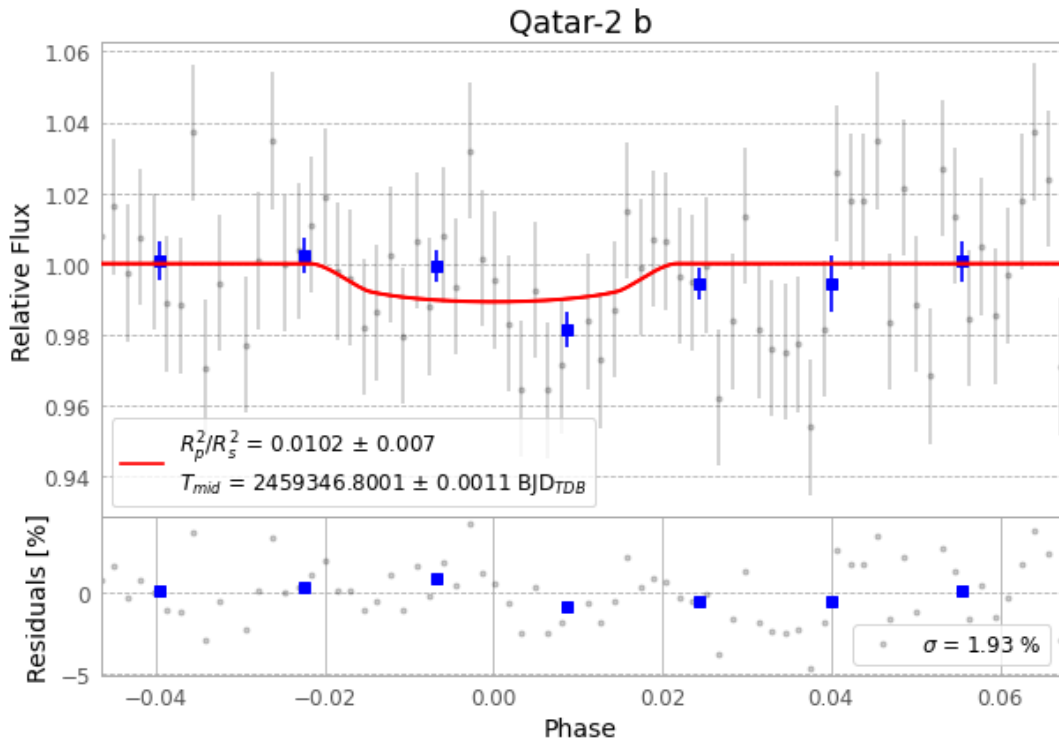


Figure 1 — FinalLightCurve_Qatar-2 b_2021-05-11.png (SHA-256 b0c91b56054b40bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [513, 27], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6a0c2940fc1093f8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), Qatar-2210512053312.FITS → Qatar-2210512091213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-210512.log (dba7394853b9...), FinalLightCurve_Qatar-2 b_2021-05-11.png (b0c91b56054b...), FinalLightCurve_Qatar-2 b_2021-05-11.csv (6f6f0fec9c8a...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 02:31Z.